Volatility Forecasting with GARCH

EXECUTIVE SUMMARY

Problem

Financial returns exhibit volatility clustering: large movements tend to follow large movements, while quiet periods persist. This project asks whether GARCH improves one-day-ahead variance forecasts for SPY relative to rolling variance and exponentially weighted moving average (EWMA), and whether Student's t shocks improve on Gaussian shocks.

Methodology

We implement the forecasting recursions and likelihood functions in Python using SPY adjusted daily closing prices from Yahoo Finance. Daily log returns are split chronologically into training (2010-2015), validation (2016-2018), and testing (2019-2025). Validation selects a 60-day rolling window and an EWMA decay of 0.94. Gaussian GARCH(1,1) and variance-one Student's t GARCH(1,1) are fitted by maximum likelihood, then refitted on 2010-2018 before testing. Parameters remain fixed during testing; variance forecasts update daily using past returns.

All four models are evaluated on the same 1,760 test days. QLIKE is the primary loss and mean squared error (MSE) is secondary; both compare predicted daily variance with observed squared shocks around a common pretest mean. Weekly averages are used only for visualization.

Principal findings

Gaussian GARCH achieves the lowest QLIKE and MSE, followed by Student's t GARCH, EWMA, and rolling variance. Lower values indicate better performance for both losses.

Model	QLIKE	MSE
Gaussian GARCH	-8.259572	2.849544e-07
Student's t GARCH	-8.256195	2.906149e-07
EWMA	-8.193821	3.276058e-07
Rolling	-8.054142	3.893375e-07

GARCH's separate shock-response and persistence coefficients, together with mean reversion, offer plausible reasons for its advantage over the benchmarks. Student's t shocks do not improve variance forecast accuracy in this sample; this does not establish that Gaussian shocks better describe return tails.

Important limitations

Conditional variance is unobserved, so squared shocks are noisy evaluation targets. Results cover one asset and one test period, and no statistical significance test has been performed. The Student's t extension was added after earlier test results were inspected, making that comparison exploratory. Historical adjusted prices may be revised and dependencies are not pinned. These findings concern variance forecasting, not trading profitability or option pricing.